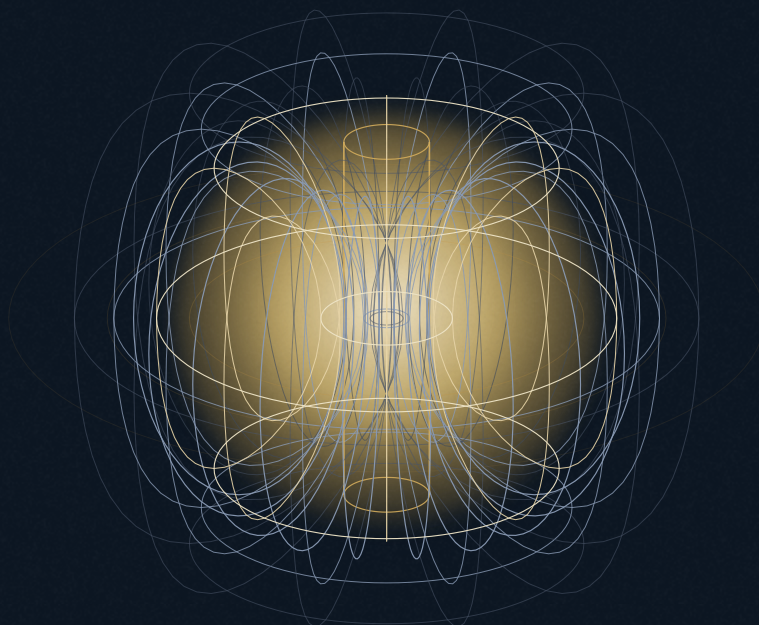




KRONOS FUSION ENERGY



PAPER 2.18 OF THE 2026 KRONOS SET

The Tritium Fuel Cycle

A Breeding Ladder, Not a Number

The breeder's product is tritium, and its credibility rests on an honest mass balance — developed here as a confirmed ladder.

Burn 4.76 kg-T/fpy · net TBR to 1.42 · ~2.0 kg-T/yr/unit

Explore it live — [3-D model](#) · [physics validator](#) · [film series](#)

THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TITLE	The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study
AUTHOR	Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy
EDITION	2026 Combined Editorial · first issue
COMPANION WORKS	The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop.
NAMING	Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.
STATUS	Conceptual design & simulation study. Nothing herein is a construction commitment.



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the *Apparatus* — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

	HYPERION	AEGIS	METROVOLT
Role	Strategic-isotope foundry	Defense installation power	Campus power
Machine	Spherical tokamak	Tandem mirror	Tandem mirror
Fuel	Deuterium–tritium	Deuterium–helium-3	Deuterium–helium-3
Delivers	Tritium · ³ He · 14 MeV n	Resilient installation power	Firm campus power
Physics bar	Fusion gain only	Closure (gates named)	Closure + lunar ³ He
In the fleet	Breeds the fuel	Proves the generator	Commercial destination

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

ABSTRACT

The Tritium Fuel Cycle — A Breeding Ladder, Not a Number

A breeder's product is tritium, and its credibility rests on an honest mass balance. The per-unit burn rate is fixed by the fusion power at 4.76 kg-T per full-power-year, and the net surplus is a confirmed *ladder*: net TBR ~1.06 (literal recipe), ~1.19 (solid-beryllium layer), and 1.42 (advanced blanket, ~2.0 kg-T/yr per unit) — the last the confirmed per-unit ceiling, above which the spherical-tokamak centrepost caps the inboard coverage. A single advanced unit is a tritium exporter; a national-scale 4 kg-T/yr supply is a *fleet* property, never a single-unit claim.

Companion deposit (code & data, reproducible under CC BY 4.0): DOI [10.5281/zenodo.22132123](https://doi.org/10.5281/zenodo.22132123) (v2 [10.5281/zenodo.21795620](https://doi.org/10.5281/zenodo.21795620)) · Zenodo community *kronos_fusion_energy*.

Explore it live: [interactive 3-D model](#) · [run the physics validator](#) · [companion film series](#) (the deposited solvers, in the browser).

Keywords: [tritium](#) [breeding ratio](#) [fuel cycle](#) [beryllium multiplier](#) [fleet aggregation](#)

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NOMENCLATURE

Symbols, units, and the names of things

Q	plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$
Q_E	engineering gain, net electric out / recirculating in
H_{98}	confinement enhancement over the IPB98(y,2) scaling
Z_{eff}	effective ion charge
c_{ee}	electron–electron bremsstrahlung leading coefficient, 2.120022 (exact)
τ_E	energy confinement time
I_p	plasma current
R_0, a	major radius, minor radius
κ, δ	elongation, triangularity
β_N	normalized plasma beta (Troyon-normalized)
TBR	tritium breeding ratio
DEC	direct energy conversion
CF	plant capacity factor (availability)
$FOAK/NOAK/BOAK$	first / n th / best of a kind
<i>Hyperion</i>	the ST breeder (internal: Mode D2)
<i>Aegis</i>	the generator, defense/installation housing (internal: Mode M)
<i>MetroVolt</i>	the generator, data-center housing (Mode M)

§ 1

A breeder is only as credible as its mass balance

A fusion breeder's product is tritium, and the temptation, historically, has been to quote a single flattering breeding ratio and move on. We do the opposite. We develop the tritium fuel cycle as a confirmed *ladder* of design points, derive the burn rate from first principles, and separate what a single unit delivers from what a fleet delivers — because conflating the two is how prior breeder concepts overstated themselves.

THE CLAIM, STATED HONESTLY

A single advanced unit is a tritium **exporter** at the $\sim 2.0 \text{ kg-T yr}^{-1}$ scale. A national-scale **4 kg-T yr⁻¹** supply is a *fleet* property — never a single-unit claim, because the spherical-tokamak centrepost caps the per-unit ceiling at net TBR 1.42.

The burn rate, and the surplus

Each 17.59 MeV deuterium–tritium fusion consumes one triton, so the per-unit burn rate follows directly from the fusion power: $\Lambda_T = 4.76 \text{ kg-T per full-power-year}$ at 85.04 MW. The net surplus available for export is then $S = (TBR - 1) \Lambda_T$. The breeding ratio, not the burn rate, is therefore the design lever — and it is a lever, not a constant.

DERIVED FROM P_{FUS}

The breeding ladder

Three-dimensional OpenMC neutronics on ENDF/B-VIII.0 place the ladder. The literal frozen recipe gives net TBR ~1.06 and a surplus of 0.29 kg-T yr⁻¹; a dedicated 5 cm solid-beryllium multiplier raises it to ~1.19 and 0.9; and the advanced blanket — a void-reduced breeder with a solid-beryllium multiplier and a steel reflector — reaches net TBR **1.42** and about **2.0 kg-T yr⁻¹ per unit**.

This TBR 1.42 is the confirmed per-unit ceiling across twelve blanket configurations; a lithium–lead alternative was explored and does not beat it, and the result reproduces across an independent nuclear-data evaluation to under 0.3%. CROSS-LIBRARY REPRODUCED

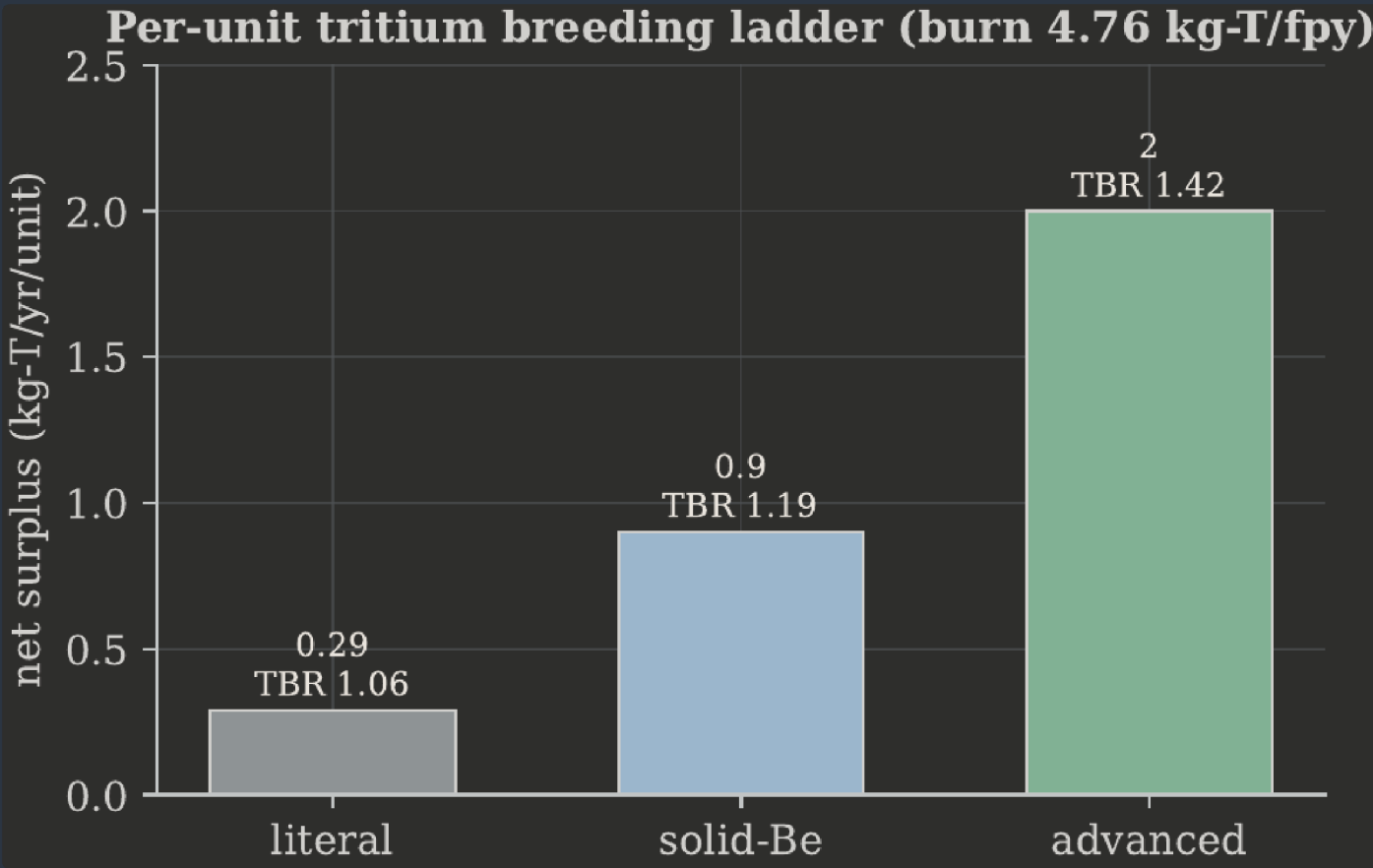
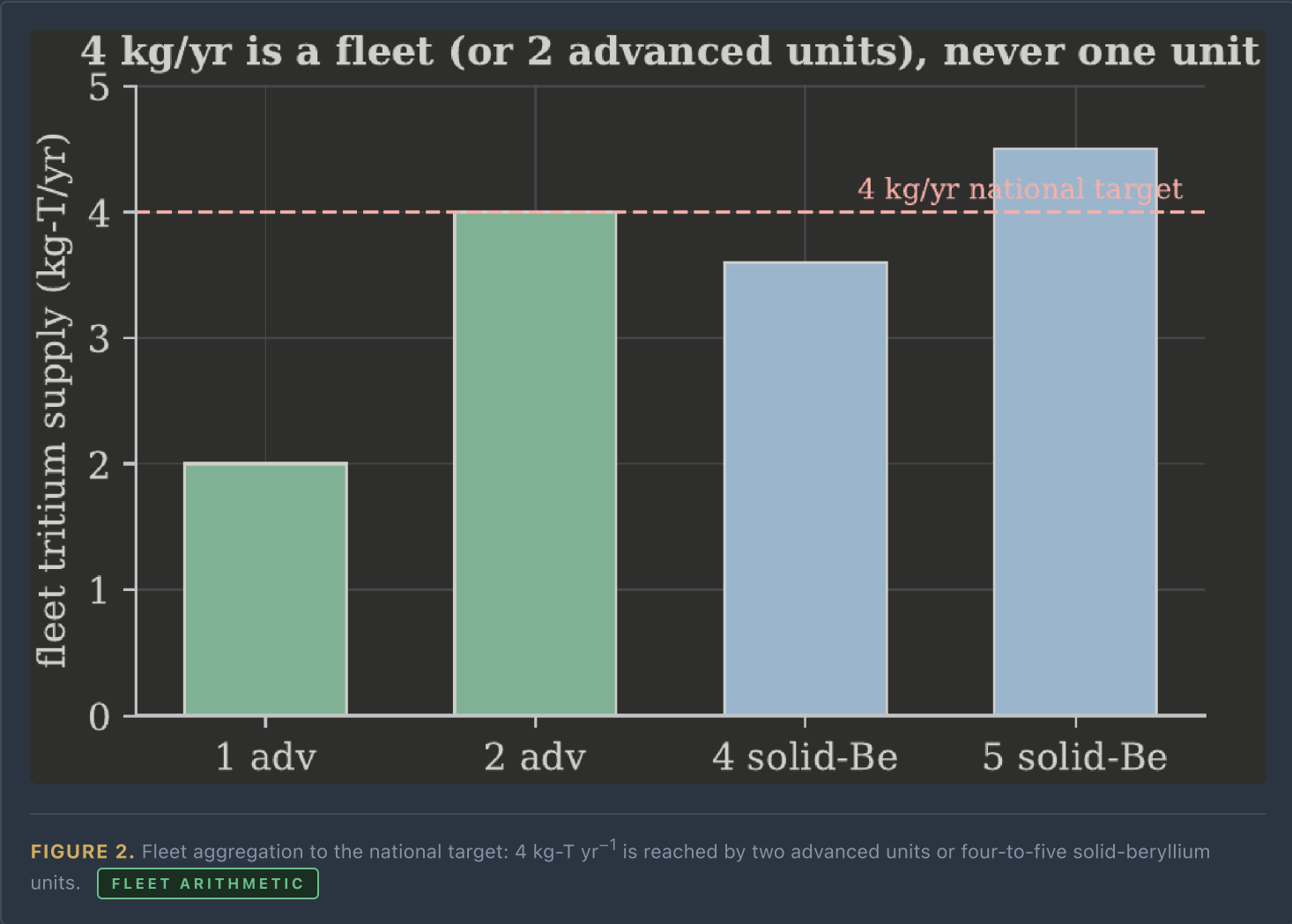


FIGURE 1. The per-unit tritium breeding ladder: net surplus at the literal, solid-beryllium, and advanced blankets, with the corresponding net breeding ratio. OPENMC / ENDF-B-VIII.0

Per unit versus fleet

The distinction is load-bearing. A single advanced unit exports tritium at the $\sim 2.0 \text{ kg-T yr}^{-1}$ scale. A national-scale 4 kg-T yr^{-1} supply is reached by two advanced units, or four-to-five solid-beryllium units — and the co-produced helium-3 aggregates the same way, to $3.6\text{-}4.5 \text{ kg yr}^{-1}$ across the fleet. We never claim 4 kg-T yr^{-1} from one unit.



The closed loop

The fuel cycle closes through direct internal recycling of unburned tritium from the exhaust, blanket tritium extraction, detritiation of process streams, and permeation control at the blanket-coolant boundary. Direct internal recycling minimizes the standing inventory — and the standing inventory, not the breeding ratio, is what the safety case turns on — by returning unburned fuel promptly rather than through a large buffered store.

Helium-3, bred by the decay of surplus tritium at equilibrium, is a co-product of that inventory rather than an independent channel. The standing-inventory value that sets the accident source term is computed in the fuel-cycle model and carried to the safety companion — flagged as the governing safety quantity, not asserted here.

COMPUTED DOWNSTREAM

APPARATUS

Portfolio, People & Record

The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.

1

GRANTED PATENT

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INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

US 12,009,112	High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550	GRANTED
US 17/878,507	Core device / method (KRONO-001A) · utility	PENDING

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

64/124,209	Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026	FILED
64/124,220	Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026	FILED
64/115,167	KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026	FILED
64/005,440	Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026	FILED
64/105,530	Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026	FILED
64/128,097	Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop	FILED

TRADEMARK & ORIGIN

FTK-98801894	Kronos Fusion Energy — trademark application	FILED
KR-2022-★	Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022	PRIORITY

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any $Q > 1$ milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

THE TEAM

Founding partners, board, advisors & auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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Legal counsel is retained; those roles are held on the internal roster and are not listed here.

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COLOPHON

How this document was made

The Kronos Fleet — Combined Editorial, 2026 is set in **Fraunces** (display), **Gelasio** (text), and **IBM Plex Mono** (data & equations), carried forward from the Kronos editorial design system.

The figures are rendered at 300 dpi from the deposited generator scripts; the document is composed as a single self-contained file with embedded fonts and imagery, paginated to US Letter, and rendered to PDF through a headless Chromium engine in matched light and dark editions.

Every headline quantity carries an evidence class; withdrawn and restated values are marked as such. Nothing herein is frozen beyond the founder-approved Tier-A set.

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Numbers and their classes. Quantities are reported with evidence classes and case labels; modelled, assumed, and requirement-class values are identified as such and are not to be quoted without their labels.

Public edition. This edition carries the physics and design only; all financial, funding, and defense-commercial content has been removed for public distribution. The scientific text corresponds to the arXiv and journal submissions.

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THE 2026 KRONOS SET

One programme, in five papers

This editorial is one of five that together describe the Kronos fusion programme — a breeder that funds a generator, and the magnet, conversion, and control science that enable both. Each is published open-access with a reproducible companion deposit.

PAPER I	Hyperion — The Tritium Foundry · 10.5281/zenodo.21746157
PAPER II	Aegis & MetroVolt — The Generator · 10.5281/zenodo.21746479
PAPER III	High-Field REBCO Magnets & Tape · 10.5281/zenodo.21842514
PAPER IV	Direct Energy Conversion · 10.5281/zenodo.21842864
PAPER V	AI, ML & Quantum Control · 10.5281/zenodo.21842371

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